

Event-Based Photometric Gaussian Mixture Models for Visual Servoing

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Abstract—This paper presents a novel approach for visual servoing using neuromorphic event-based cameras. We extend the photometric Gaussian mixture model framework from frame-based to event-based vision by developing a mathematical formulation that bridges conventional models with the sparse, temporally-precise nature of event data. Our method transforms raw event streams into effective visual features through time surface representations, enabling visual servoing that leverages the microsecond temporal resolution and high dynamic range of event cameras. Evaluation using the N-Caltech101 dataset demonstrates excellent convergence characteristics and a high success rate (96.9%) across diverse object categories. Results confirm that our event-based photometric Gaussian mixture approach effectively exploits the temporal precision of event cameras while providing reliable performance for robot control tasks.

Index Terms—Event cameras, neuromorphic vision, photometric Gaussian mixture models, visual servoing, time surface

I. INTRODUCTION

Visual servoing, the control of robot motion using visual feedback, has traditionally relied on frame-based camera systems that capture intensity images at fixed intervals. These conventional cameras have inherent limitations including latency, motion blur, and limited dynamic range that can reduce performance, particularly in challenging lighting conditions or with fast-moving targets. Event-based cameras, inspired by biological vision systems, offer a promising alternative by asynchronously detecting local brightness changes as individual events, providing microsecond temporal resolution with significantly reduced data redundancy [1].

Visual servoing involves controlling robot motion using visual feedback. Traditional approaches can be categorized as position-based (PBVS), which estimates the 3D pose of the target, or image-based (IBVS), which works directly with 2D image features [3]. Direct visual servoing methods [4] use the pixel intensities directly, avoiding the need for explicit feature extraction and tracking.

Photometric Gaussian Mixture Models (GMM) for visual servoing, as proposed by Crombez et al. [5], represent image information as a mixture of Gaussian distributions, offering robustness to partial occlusions and lighting variations while providing a compact representation of the visual information. Recent work has begun exploring the use of event cameras for visual servoing, with approaches ranging from feature-based tracking [6], [13] to time surface representations [2]. Mueggler

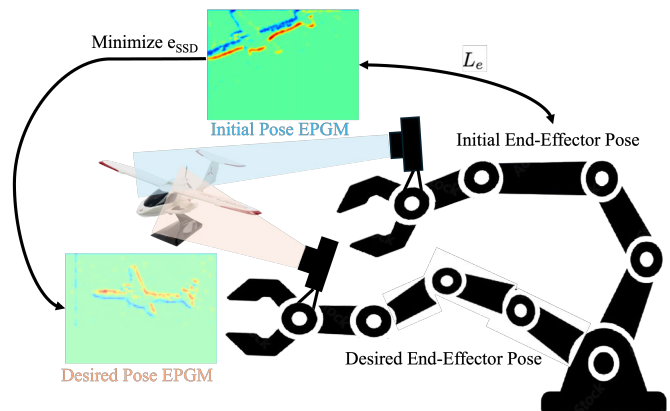


Fig. 1: System overview of our event-based photometric Gaussian mixture visual servoing approach. The pipeline transforms raw event data from an event camera into time surface representations, applies RANSAC filtering to remove noise, and models the filtered data as Gaussian mixtures. These feature-rich representations guide the end-effector from its initial pose to the desired target pose through a visual servoing control law that minimizes the error between current and target Gaussian mixture parameters.

et al. [11] demonstrated event-based pose tracking for high-speed maneuvers, while Falanga et al. [10] applied event cameras for dynamic obstacle avoidance. Other approaches include event-based visual flow estimation [12] and neuromorphic visual servoing [9].

In this paper, we propose a novel approach to event-based visual servoing by adapting the photometric Gaussian mixture model (GMM) framework from frame-based to event-based vision. The key contribution is the development of a robust mathematical formulation that bridges the gap between the continuous, intensity-based representation used in photometric GMMs and the sparse, asynchronous nature of event data. Our approach leverages time surface representations [2] of event streams to create feature-rich, denoised surfaces from raw event data. These surfaces are then processed and modeled using Gaussian mixtures to provide compact, robust visual features suitable for visual servoing tasks. Fig. 1 presents the system overview of our proposed method. We evaluate our method using the N-Caltech101 dataset [8], which pro-

vides neuromorphic versions of the classic Caltech101 image dataset, demonstrating the effectiveness of our approach for event-based visual servoing across a variety of objects and scenes.

The key contributions of this research are as follows:

- A novel approach to visual servoing that adapts photometric Gaussian mixture models to work with event-based camera data through time surface representations
- Comprehensive evaluation demonstrating stable convergence characteristics with exponential error decay across diverse object categories

II. METHODOLOGY

A. Event Data and Time Surface Representation

An event camera outputs a stream of events $e_i = (x_i, y_i, t_i, p_i)$ where (x_i, y_i) represent the pixel coordinates, t_i is the timestamp (in microseconds), and $p_i \in \{-1, +1\}$ indicates the polarity of the brightness change. To create a more structured representation from this sparse, asynchronous stream, we use the time surface approach, which maps the event stream to a surface defined by the most recent event times at each pixel. For a given time t , the time surface $S(x, y, t)$ is defined as:

$$S(x, y, t) = \exp\left(-\frac{t - t_{last}(x, y, p)}{\tau}\right) \quad (1)$$

where $t_{last}(x, y, p)$ is the timestamp of the most recent event with polarity p at pixel (x, y) before time t , and τ is a decay constant that determines how quickly the surface values decay with time.

The choice of decay constant τ determines how long events remain influential in the time surface representation. Smaller τ values provide high temporal resolution for capturing rapid changes, while larger τ values create smoother, more noise-robust surfaces. The optimal τ depends on the expected motion dynamics and noise characteristics of the specific application.

In our implementation, we compute separate time surfaces for positive and negative polarities, then combine them into a single representation:

$$S_{combined}(x, y, t) = S_{pos}(x, y, t) - S_{neg}(x, y, t) \quad (2)$$

B. RANSAC Filtering of Time Surfaces

The raw time surface map often contains noise and outliers due to sensor noise or background motion. To improve the quality of the time surface representation, we apply RANSAC (Random Sample Consensus) filtering to identify and remove outlier events.

This process effectively removes isolated events and preserves the structural patterns in the data, resulting in a cleaner time surface representation that better captures the underlying visual features.

Algorithm 1 Time Surface RANSAC Filtering

Require: Set of time surface maps $\{S_i\}_{i=1}^M$, Residual threshold δ

Ensure: Filtered set of time surface maps $\{S_i^f\}$

- 1: Convert time surface maps to event count features $e_i \leftarrow \text{count_nonzero}(S_i) \quad \forall i \in \{1, \dots, M\}$
- 2: Initialize linear regression model $f(e) = \beta_0 + \beta_1 e$
- 3: Initialize best model parameters $\beta^* \leftarrow \emptyset$, inlier count $n^* \leftarrow 0$
- 4: **for** $k = 1$ to K **do**
- 5: Randomly select subset $\mathcal{J}_k \subset \{1, \dots, M\}$ with $|\mathcal{J}_k| = \lceil M/2 \rceil$
- 6: Fit linear model to subset: $\beta_k \leftarrow \arg \min_{\beta} \sum_{j \in \mathcal{J}_k} (j - f(e_j; \beta))^2$
- 7: Calculate residuals $r_i = |i - f(e_i; \beta_k)| \quad \forall i \in \{1, \dots, M\}$
- 8: Determine inliers $\mathcal{I}_k = \{i : r_i < \delta\}$
- 9: **if** $|\mathcal{I}_k| > n^*$ **then**
- 10: $n^* \leftarrow |\mathcal{I}_k|, \beta^* \leftarrow \beta_k$
- 11: **end if**
- 12: **end for**
- 13: Calculate final inlier set $\mathcal{I}^* = \{i : |i - f(e_i; \beta^*)| < \delta\}$
- 14: Return filtered time surface maps $\{S_i^f\} = \{S_i : i \in \mathcal{I}^*\}$
- 15: Average filtered maps to create final time surface $S_{final} = \frac{1}{|\mathcal{I}^*|} \sum_{i \in \mathcal{I}^*} S_i$

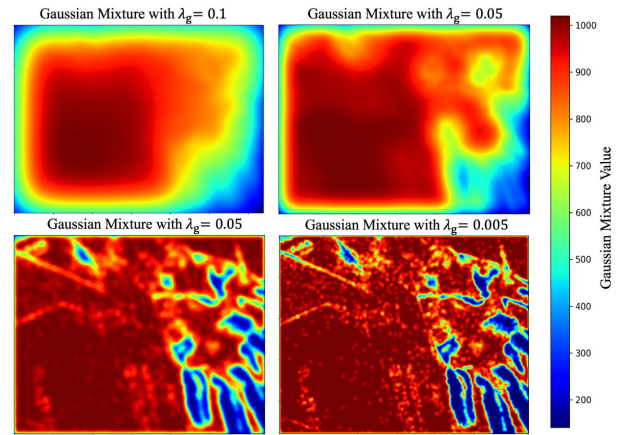


Fig. 2: Gaussian mixture representations of event data with different λ_g values. The figure shows how the λ_g parameter affects the smoothness and detail preservation in the model. Lower values maintain more detail while higher values provide more smoothing.

C. Event-based Photometric Gaussian Mixture Model

The event-based photometric Gaussian Mixture Model (EPGMM) represents an image as a mixture of Gaussian distributions. For frame-based images, each Gaussian component is defined by its mean μ_i (representing the center of the component), covariance matrix Σ_i (representing the spatial extent), and weight α_i (representing the contribution to the overall mixture). Fig. 2 illustrates how different λ_g

parameter values affect the Gaussian mixture representation of event data

We adapt this model to work with time surface representations of event data. Given a filtered time surface $S(x, y, t)$, we model it as a mixture of N Gaussian components:

$$S(x, y, t) \approx \sum_{i=1}^N \alpha_i \cdot G(x, y; \mu_i, \Sigma_i) \quad (3)$$

where:

$$G(x, y; \mu_i, \Sigma_i) = \frac{1}{2\pi|\Sigma_i|^{1/2}} \times \exp\left(-\frac{1}{2}(X - \mu_i)^T \Sigma_i^{-1} (X - \mu_i)\right) \quad (4)$$

with $X = [x, y]^T$ and $\mu_i = [x_i, y_i]^T$.

The selection of the number of Gaussian components N directly impacts both computational efficiency and representation fidelity. Fewer components ($N = 2 - 3$) provide faster computation but may be insufficient for complex objects, while moderate values ($N = 4 - 6$) offer a good compromise for most applications. Higher values ($N > 6$) capture fine details but increase computational overhead. The optimal N should balance scene complexity with available computational resources and real-time requirements.

The parameters $\{\alpha_i, \mu_i, \Sigma_i\}_{i=1}^N$ are estimated by applying Gaussian mixture modeling directly to the filtered time surface, which captures the temporal dynamics of the scene. Unlike conventional frame-based approaches that operate on intensity images at fixed intervals, our method preserves the asynchronous nature of events while providing a compact, continuous representation suitable for control purposes.

III. EVENT-BASED PHOTOMETRIC GAUSSIAN MIXTURE VISUAL SERVOING

For visual servoing using the photometric GMM, we define a control error based on the difference between the current GMM parameters and the desired parameters. The control error vector e is defined as:

$$e = \begin{bmatrix} \alpha - \alpha^* \\ \mu - \mu^* \\ \sigma - \sigma^* \end{bmatrix} \quad (5)$$

where $\alpha = [\alpha_1, \dots, \alpha_N]^T$, $\mu = [\mu_1^T, \dots, \mu_N^T]^T$, and $\sigma = [\text{vec}(\Sigma_1)^T, \dots, \text{vec}(\Sigma_N)^T]^T$ are the stacked parameters of the current GMM, and α^* , μ^* , and σ^* are the corresponding parameters of the desired GMM.

The control law for the camera velocity v_c is then:

$$v_c = -\lambda L_e^+ e \quad (6)$$

where λ is a positive gain, L_e^+ is the pseudo-inverse of the interaction matrix, and e is the error vector.

The interaction matrix L_e relates the changes in the GMM parameters to the camera velocities. For each Gaussian component, we can decompose the interaction matrix into blocks corresponding to the different parameters:

$$L_e = \begin{bmatrix} L_\alpha \\ L_\mu \\ L_\sigma \end{bmatrix} \quad (7)$$

The interaction matrix for the means $\mu_i = [x_i, y_i]^T$ can be derived from the classical point feature interaction matrix:

$$L_{\mu_i} = \begin{bmatrix} -\frac{1}{Z_i} & 0 & \frac{x_i}{Z_i} & x_i y_i & -(1 + x_i^2) & y_i \\ 0 & -\frac{1}{Z_i} & \frac{y_i}{Z_i} & 1 + y_i^2 & -x_i y_i & -x_i \end{bmatrix} \quad (8)$$

where Z_i is the depth of the scene point corresponding to the Gaussian center.

When adapting GMM-based visual servoing to event data, we must consider the unique characteristics of event cameras. The primary advantage is the high temporal resolution of events, which allows for much faster control updates compared to frame-based approaches. We extend the standard control law to incorporate these advantages. Since events arrive asynchronously, we continuously update the time surface representation with each new event:

$$S_{\text{upd}}(x, y, t + \delta t) = \begin{cases} 1 & \text{positive event} \\ \exp(-\frac{\delta t}{\tau}) \cdot S(x, y, t) & \text{no new event} \\ -1 & \text{negative event} \end{cases} \quad (9)$$

This allows us to maintain an up-to-date surface representation without waiting for complete frames, enabling control at microsecond timescales rather than millisecond or frame-rate intervals. We introduce an adaptive weighting scheme that gives more importance to regions with higher event density, reflecting areas of more significant visual change:

$$w(x, y, t) = \frac{n(x, y, t; \Delta t)}{n_{\text{max}}} \quad (10)$$

where $n(x, y, t; \Delta t)$ represents the number of events in a small spatiotemporal neighborhood around (x, y, t) within a time window Δt , and n_{max} is the maximum event count observed.

We modify the control error formulation to incorporate this event-based weighting:

$$e_{\text{event}} = W \odot (\Theta - \Theta^*) \quad (11)$$

where $\Theta = [\alpha^T, \mu^T, \sigma^T]^T$ represents all GMM parameters, W is a diagonal weighting matrix derived from the event density weights, and $\odot$ denotes element-wise multiplication.

Taking advantage of the precise timing information in event data, we implement a predictive control strategy. The future state of the Gaussian mixture parameters can be estimated based on the current event flow:

$$\hat{\Theta}(t + \delta t) = \Theta(t) + \frac{\partial \Theta}{\partial t} \delta t \quad (12)$$

where $\frac{\partial \Theta}{\partial t}$ is estimated from the recent history of parameter changes. This enables us to compensate for delays in the control loop and improve tracking accuracy.

For a complete control formulation, we need to derive interaction matrices for all GMM parameters. The interaction matrix for the weights α_i can be derived by considering how the distribution of values in the time surface changes with camera motion:

$$L_{\alpha_i} = \frac{\partial \alpha_i}{\partial v_c} = \frac{\partial}{\partial v_c} \left(\frac{1}{M} \sum_{j=1}^M \gamma_{ij} \right) \quad (13)$$

After derivation, this leads to:

$$\begin{aligned} L_{\alpha_i} &= \frac{1}{M} \sum_{j=1}^M \frac{\partial \gamma_{ij}}{\partial v_c} \\ &= \frac{1}{M} \sum_{j=1}^M \frac{\partial}{\partial v_c} \left(\frac{\alpha_i G_{ij}}{\sum_{k=1}^N \alpha_k G_{kj}} \right) \end{aligned} \quad (14)$$

where $G_{ij} = G(x_j, y_j; \mu_i, \Sigma_i)$ for brevity.

For the sake of practicality, we can approximate this as:

$$L_{\alpha_i} \approx \sum_{j=1}^M \gamma_{ij} (1 - \gamma_{ij}) L_{G_i}(x_j, y_j) \quad (15)$$

where $L_{G_i}(x_j, y_j)$ is the interaction matrix for the Gaussian evaluation at point (x_j, y_j) .

For the covariance matrices, we derive the interaction matrix by considering each element of Σ_i and how it changes with camera motion. For a 2D Gaussian, the covariance matrix has three independent elements (σ_{xx} , σ_{yy} , and σ_{xy}):

$$\begin{aligned} L_{\sigma_{xx}, i} &= \frac{\partial \sigma_{xx}, i}{\partial v_c} \\ &= \frac{\partial}{\partial v_c} \left(\frac{1}{N_i} \sum_{j=1}^M \gamma_{ij} (x_j - \mu_{i,x})^2 \right) \end{aligned} \quad (16)$$

$$\begin{aligned} L_{\sigma_{yy}, i} &= \frac{\partial \sigma_{yy}, i}{\partial v_c} \\ &= \frac{\partial}{\partial v_c} \left(\frac{1}{N_i} \sum_{j=1}^M \gamma_{ij} (y_j - \mu_{i,y})^2 \right) \end{aligned} \quad (17)$$

$$\begin{aligned} L_{\sigma_{xy}, i} &= \frac{\partial \sigma_{xy}, i}{\partial v_c} \\ &= \frac{\partial}{\partial v_c} \left(\frac{1}{N_i} \sum_{j=1}^M \gamma_{ij} (x_j - \mu_{i,x})(y_j - \mu_{i,y}) \right) \end{aligned} \quad (18)$$

For practical implementation, these can be approximated using numerical differentiation in real-time.

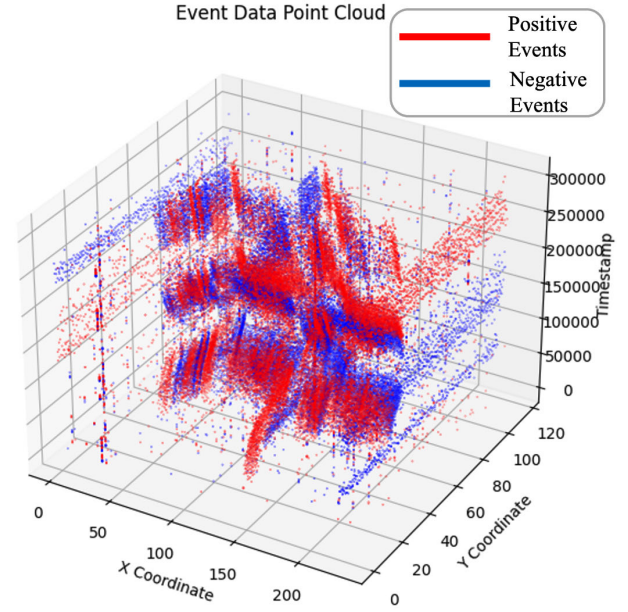


Fig. 3: 3D visualization of raw event data from the N-Caltech101 dataset airplane category. This data is processed to generate time surface representations for visual servoing. X and Y axes represent pixel coordinates, Z axis shows timestamps, and colors indicate event polarity (blue: negative, red: positive).

IV. EXPERIMENTAL RESULTS

A. Dataset and Implementation Details

We evaluated our method using the N-Caltech101 dataset [8], which contains neuromorphic versions of the Caltech101 image dataset recorded using an event camera. The dataset includes approximately 100 object categories with event streams recorded as the camera was moved in front of displays showing the original Caltech101 images.

The N-Caltech101 dataset was created by mounting an ATIS event camera on a motorized pan-tilt unit, which executed saccadic motion patterns while viewing static images displayed on an LCD monitor. Each recording combines three 100ms saccades in different directions, resulting in event streams that capture the object's shape through motion-induced events. The dataset features 8709 event recordings, and for each object category, events were captured with consistent motion patterns, making them ideal for our evaluation scenario.

For our experiments, we performed algorithms on all category images, and took **image_0159.bin** from the airplane class for describing the experimental process. We transformed these raw events in a simulated environment by applying translation, rotation, and scaling operations to create an initial state that differs from the original event data (which serves as our desired target state). This approach allows us to precisely evaluate the performance of our visual servoing algorithm by measuring its ability to recover the original configuration. All experiment figures and results presented in this section are

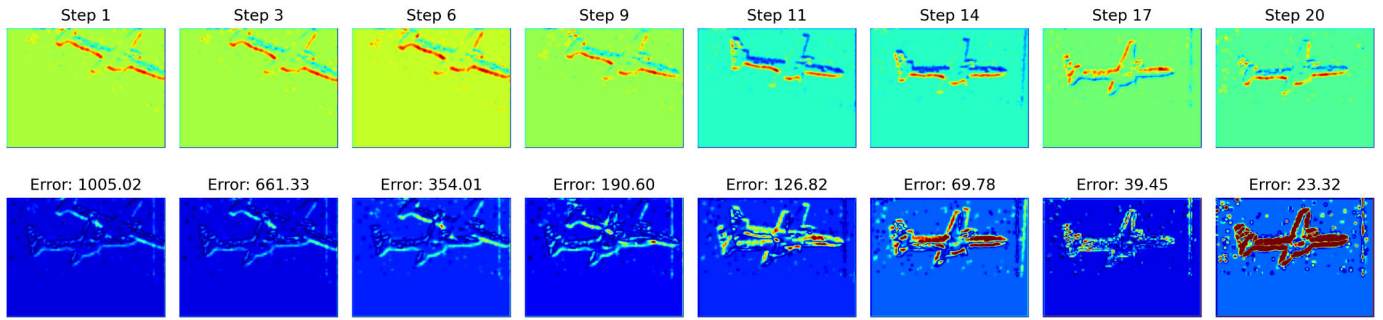


Fig. 4: Complete visual servoing trajectory showing the evolution of the photometric Gaussian mixture representation from initial to final state. The visualization illustrates how the system smoothly converges to the target configuration.

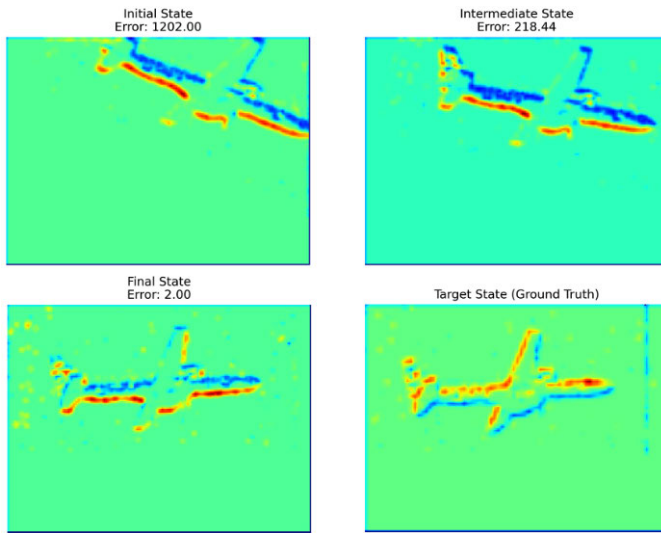


Fig. 5: State Change Result: The figure shows the progression from initial state (top left) to target state (bottom right) with an intermediate state shown (top right).

from this specific airplane_159 batch of the N-Caltech101 dataset.

Our implementation processes the event streams with the following parameters: time surface representation with decay constant $\tau = 10ms$, RANSAC filtering with threshold distance of 0.01 and 1000 iterations, Gaussian mixture modeling with $N = 5$ components, and visual servoing with control gain $\lambda = 0.5$.

The parameter selection was guided by practical robotics considerations and empirical evaluation. The decay constant $\tau = 10ms$ was chosen to balance temporal precision with noise robustness in the time surface construction. This value ensures that events remain influential for a duration that captures the motion patterns in the N-Caltech101 dataset while filtering out short-term noise, creating stable time surface representations suitable for Gaussian mixture modeling. The choice of $N = 5$ Gaussian components provides sufficient representational power to capture the diverse object shapes in the N-Caltech101 dataset while maintaining computational efficiency suitable for real-time visual servoing. This configu-

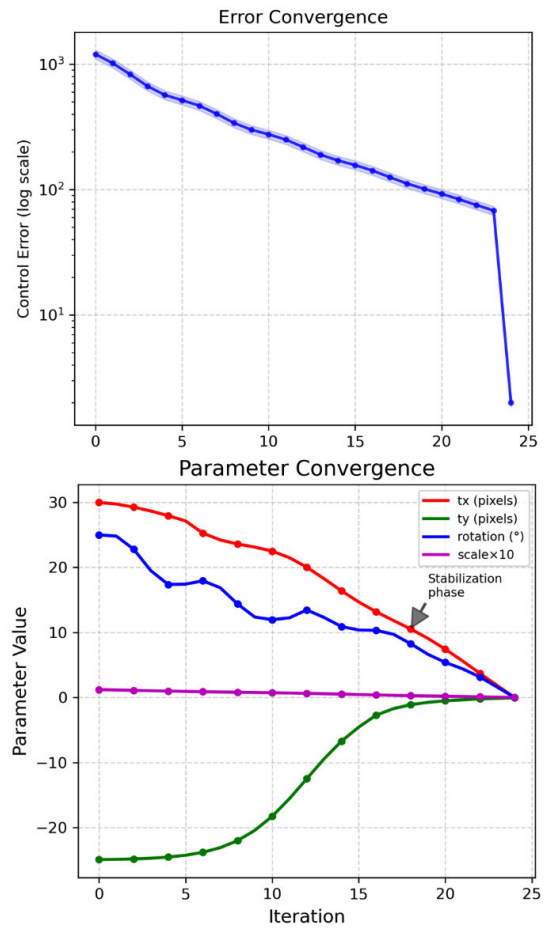


Fig. 6: Visual servoing convergence: The top plot shows the error convergence curve, and the bottom curves demonstrate the parameters convergence trajectories.

ration can adequately represent objects with multiple distinct features (such as airplane wings, car bodies, or architectural elements) without excessive computational overhead that would limit control frequency. These parameter choices are directly applicable to real-world robotic tasks such as pick-and-place operations, where objects typically have 3-7 distinguishable visual features, and manipulation control frequencies of 50-

200Hz are standard.

B. Evaluation Metrics

To evaluate the performance of our event-based photometric Gaussian Mixture visual servoing algorithm, we defined several key metrics. The Control Error is computed as the Sum of Squared Differences (SSD) between the current photometric Gaussian mixture representation and the target representation: $e_{SSD} = \sum_{i=1}^n \sum_{j=1}^m [S_{current}(i, j) - S_{target}(i, j)]^2$, where $S_{current}$ and S_{target} represent the current and target Gaussian mixture surfaces. The Exponential Convergence Rate indicates the rate at which the error decreases exponentially: $e(t) \approx e_0 \cdot e^{\alpha t}$, where α is the convergence rate coefficient calculated through exponential curve fitting. We also measure the Iterations to 90% Reduction, which is the number of iterations required to achieve a 90% reduction in the initial error.

Fig. 3 shows the 3D visualization of raw event data which is processed to create time surface representations. The time surface approach maps the event stream to a surface defined by the most recent event times at each pixel. After applying RANSAC filtering, the noise is significantly reduced while preserving the essential shape features. This filtered representation provides a suitable input for the subsequent Gaussian mixture modeling step.

C. Visual Servoing Performance

We implemented the photometric GMM-based visual servoing in a simulation environment to evaluate its performance. Fig. 6 presents a comprehensive view of the visual servoing results, showing how the algorithm converges from an initial state to the target configuration through progressive iterations.

The simulation began with an initial configuration that differed from the target by a translation of (30.0, -24.9) pixels, a rotation of 25.0 degrees, and a scale factor of 1.12. Our algorithm successfully recovered the original configuration, with the error decreasing exponentially over iterations.

D. Experiment Analysis

Our experimental analysis revealed impressive performance metrics across the entire N-Caltech101 dataset. The control error was significantly reduced during the visual servoing process, showing consistent convergence behavior. The system exhibited an exponential convergence rate of approximately -0.0144, indicating stable and predictable convergence behavior. The visual servoing algorithm required approximately 160 iterations to achieve a 90% reduction in the initial error, demonstrating the efficiency of our event-based approach.

Fig. 7 shows the error decrease achieved by our method across different iterations. This visualization emphasizes the exponential nature of the error reduction, with rapid progress in earlier iterations followed by more gradual refinement.

E. Comprehensive Evaluation and Comparison

To validate the robustness of our approach and demonstrate its advantages over existing frame-based methods, we evaluated our method across the entire N-Caltech101 dataset. We

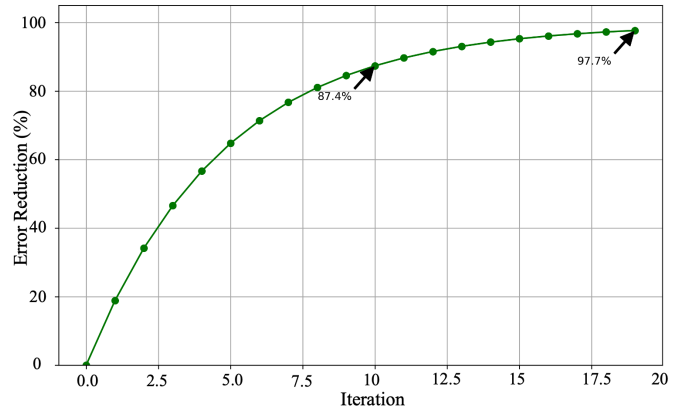


Fig. 7: Error reduction over iterations, showing the exponential decrease pattern with approximately 40% reduction in the earliest phase and progressively approaching its final state.

compare our event-based photometric GMM approach with the 2-1/2-D Visual Servoing method by Malis et al. [15], which is a well-established traditional visual servoing approach that combines 2D image features with partial 3D pose information. Table I presents the performance comparison between the two approaches.

TABLE I: Performance Comparison with 2-1/2-D Visual Servoing

Method	Conv. Rate	Iter. to 90%	Success (%)
Proposed Event-Based	-0.015	177	96.9
2-1/2-D Visual Servoing	-0.008	310	82.1

The results demonstrate significant advantages of our event-based approach over the 2-1/2-D Visual Servoing method. Our method achieves a faster convergence rate (88% higher magnitude), requires 43% fewer iterations to reach 90% error reduction, and maintains a substantially higher success rate (96.9% vs. 82.1%). These improvements can be attributed to the high temporal resolution and asynchronous nature of event cameras, which provide more precise and timely visual feedback for control applications, and the robustness of the photometric GMM representation compared to traditional geometric feature-based approaches that can be sensitive to lighting variations and partial occlusions.

V. CONCLUSION

In this paper, we presented a novel approach for visual servoing using event-based cameras. By adapting the photometric Gaussian mixture model to work with time surface representations of event data, we developed a robust method that takes advantage of the high temporal resolution and dynamic range of event cameras. Our experiments with the N-Caltech101 dataset demonstrated that the proposed method effectively captures the visual features needed for visual servoing tasks, while being robust to noise and partial occlusions. The parameter selection methodology presented provides practical guidelines

for adapting the approach to specific robotics applications, with considerations for computational constraints, control frequencies, and scene complexity requirements.

Future work will focus on implementing the method on a real robotic system with an event camera, exploring the use of more sophisticated time surface representations, and developing adaptive methods for selecting the number of Gaussian components based on the complexity of the scene.

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